

## General Qubit Hamiltonian

*From abstract matrices to rotations on the Bloch sphere*

In previous chapters, we introduced:

- Pauli matrices as the fundamental operators for two-level systems
- The qubit as a state on the Bloch sphere
- The geometric representation of pure states

In this chapter, we connect these ideas to dynamics. We show that the most general Hamiltonian of a qubit generates a rotation of the Bloch vector around some axis in space. This is the core of real quantum control and quantum computing.

### General form of a qubit Hamiltonian

Any Hermitian  $2 \times 2$  operator can be written as:

$$H = a_0 I + a_x \sigma_x + a_y \sigma_y + a_z \sigma_z = a_0 I + \vec{a} \cdot \vec{\sigma}$$

Where:

- $a_0 \in \mathbb{R}$
- $\vec{a} = (a_x, a_y, a_z) \in \mathbb{R}^3$
- $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$

This is the most general Hamiltonian for a single qubit.

The time evolution operator is:

$$U(t) = e^{-iHt/\hbar}$$

The term  $a_0 I$  only contributes a **global phase**, and therefore does not affect any observable:

$$e^{-i(a_0 I)t/\hbar} = e^{-ia_0 t/\hbar} I$$

which has no physical effect on measurement probabilities.

Therefore, the physically relevant part of the Hamiltonian is:

$$H' = \vec{a} \cdot \vec{\sigma}$$

From now on, we focus on this term.

## Geometric interpretation: $\vec{a}$ as a rotation axis

The Pauli matrices correspond to the three axes of the Bloch sphere:

- $\sigma_x \rightarrow x\text{-axis}$
- $\sigma_y \rightarrow y\text{-axis}$
- $\sigma_z \rightarrow z\text{-axis}$

The combination:

$$\vec{a} \cdot \vec{\sigma} = a_x \sigma_x + a_y \sigma_y + a_z \sigma_z$$

is therefore associated with a direction in 3D space.

We define:

$$\hat{n} = \frac{\vec{a}}{|\vec{a}|} \quad , \quad \Omega = \frac{|\vec{a}|}{\hbar}.$$

Then:

$$H' = \vec{a} \cdot \vec{\sigma} = \hbar \Omega \hat{n} \cdot \vec{\sigma}$$

The unit vector  $\hat{n}$  is the **axis of rotation** on the Bloch sphere.

The scalar  $\Omega$  is the **rotation frequency**.

## Exponential of the Hamiltonian: why it is a rotation

The time evolution operator is:

$$U(t) = e^{-iH't/\hbar} = e^{-i\Omega t \hat{n} \cdot \sigma}$$

The key algebraic fact is:

$$(\hat{n} \cdot \vec{\sigma})^2 = I.$$

for any unit vector  $\hat{n}$ .

This follows from the Pauli algebra and the fact that  $|\hat{n}| = 1$ .

Using this, the exponential can be written exactly as:

$$U(t) = \cos(\Omega t) I - i \sin(\Omega t) \hat{n} \cdot \vec{\sigma}.$$

This is the standard form of a **rotation operator** in the spin- $\frac{1}{2}$  representation.

On the Bloch sphere, it corresponds to a rotation of the Bloch vector around the axis  $\hat{n}$  with angular frequency  $\Omega$ .

## Examples: Hamiltonian proportional to $\sigma$ axis

### i. Hamiltonian proportional to $\sigma_z$

Consider:

$$H = \frac{\hbar\omega}{2}\sigma_z$$

Then:

$$U(t) = e^{-iHt/\hbar} = e^{-i(\omega t/2)\sigma_z} = \cos(\omega t/2)I - i\sin(\omega t/2)\sigma_z$$

We know:

$$\sigma_z |0\rangle = +|0\rangle \quad , \quad \sigma_z |1\rangle = -|1\rangle$$

So:

$$U(t) |0\rangle = e^{-i\omega t/2} |0\rangle \quad , \quad U(t) |1\rangle = e^{+i\omega t/2} |1\rangle$$

For a general state:

$$|\psi(0)\rangle = \alpha |0\rangle + \beta |1\rangle$$

We obtain:

$$|\psi(t)\rangle = \alpha e^{-i\omega t/2} |0\rangle + \beta e^{+i\omega t/2} |1\rangle .$$

- The populations  $|\alpha|^2$  and  $|\beta|^2$  remain constant.
- Only the **relative phase** between  $|0\rangle$  and  $|1\rangle$  changes.
- On the Bloch sphere, this is a **rotation around the z-axis**.

This type of Hamiltonian appears, for example, for a spin- $\frac{1}{2}$  in a static magnetic field along z.

**ii. Hamiltonian proportional to  $\sigma_x$**

Consider:

$$H = \frac{\hbar\Omega}{2}\sigma_x$$

Then:

$$U(t) = e^{-iHt/\hbar} = e^{-i(\Omega t/2)\sigma_x} = \cos\left(\frac{\Omega t}{2}\right)I - i\sin\left(\frac{\Omega t}{2}\right)\sigma_x$$

Using:

$$\sigma_x |0\rangle = |1\rangle \quad , \quad \sigma_x |1\rangle = |0\rangle$$

We find:

$$U(t) |0\rangle = \cos\left(\frac{\Omega t}{2}\right) |0\rangle - i\sin\left(\frac{\Omega t}{2}\right) |1\rangle$$

$$U(t) |1\rangle = \cos\left(\frac{\Omega t}{2}\right) |1\rangle - i\sin\left(\frac{\Omega t}{2}\right) |0\rangle$$

If the system starts in  $|0\rangle$ , then:

$$|\psi(t)\rangle = \cos\left(\frac{\Omega t}{2}\right) |0\rangle - i\sin\left(\frac{\Omega t}{2}\right) |1\rangle$$

The probability to find the system in  $|1\rangle$  at time  $t$  is:

$$P_{0 \rightarrow 1}(t) = |\langle 1|\psi(t)\rangle|^2 = \sin^2\left(\frac{\Omega t}{2}\right).$$

This describes **Rabi oscillations** (to be more explained later) between  $|0\rangle$  and  $|1\rangle$

On the Bloch sphere:

- the Bloch vector rotates around the x-axis
- the populations oscillate between the north and south poles

This type of Hamiltonian appears when a qubit is driven by a resonant field that couples  $|0\rangle$  and  $|1\rangle$ .

**iii. Hamiltonian proportional to  $\sigma_y$**

Consider:

$$H = \frac{\hbar\Omega}{2}\sigma_y$$

Then:

$$U(t) = e^{-iHt/\hbar} = e^{-i(\Omega t/2)\sigma_y} = \cos\left(\frac{\Omega t}{2}\right)I - i\sin\left(\frac{\Omega t}{2}\right)\sigma_y$$

Using:

$$\sigma_y|0\rangle = i|1\rangle \quad , \quad \sigma_y|1\rangle = -i|0\rangle$$

We find:

$$U(t)|0\rangle = \cos\left(\frac{\Omega t}{2}\right)|0\rangle + \sin\left(\frac{\Omega t}{2}\right)|1\rangle$$

$$U(t)|1\rangle = \cos\left(\frac{\Omega t}{2}\right)|1\rangle - \sin\left(\frac{\Omega t}{2}\right)|0\rangle$$

If the system starts in  $|0\rangle$ , then:

$$|\psi(t)\rangle = \cos\left(\frac{\Omega t}{2}\right)|0\rangle + \sin\left(\frac{\Omega t}{2}\right)|1\rangle$$

The probabilities are:

$$P(0,t) = \cos^2\left(\frac{\Omega t}{2}\right) \quad , \quad P(1,t) = \sin^2\left(\frac{\Omega t}{2}\right)$$

Again, we see oscillations between  $|0\rangle$  and  $|1\rangle$ , but with a different phase structure compared to the  $\sigma_x$  case.

On the Bloch sphere:

- The Bloch vector rotates around the y-axis
- Both amplitude and phase evolve in a specific way

In the next chapter, we will interpret these rotations explicitly as quantum gates and connect them to the standard gate set used in quantum computing.